

The background features several overlapping circles in shades of blue, purple, and pink. A pink shield icon is positioned on the left side, partially overlapping the circles.

Muhafiz

RED ZONE MAPPING AND COMMUNITY DRIVEN SUPPORT



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Introduction – Why Muhafiz?

● Urban Safety Crisis in Karachi

- 20M+ population with high crime rate
- Daily crimes: snatching, theft, harassment, gang violence
- No real-time updates on dangerous areas

● Lack of Crime Awareness

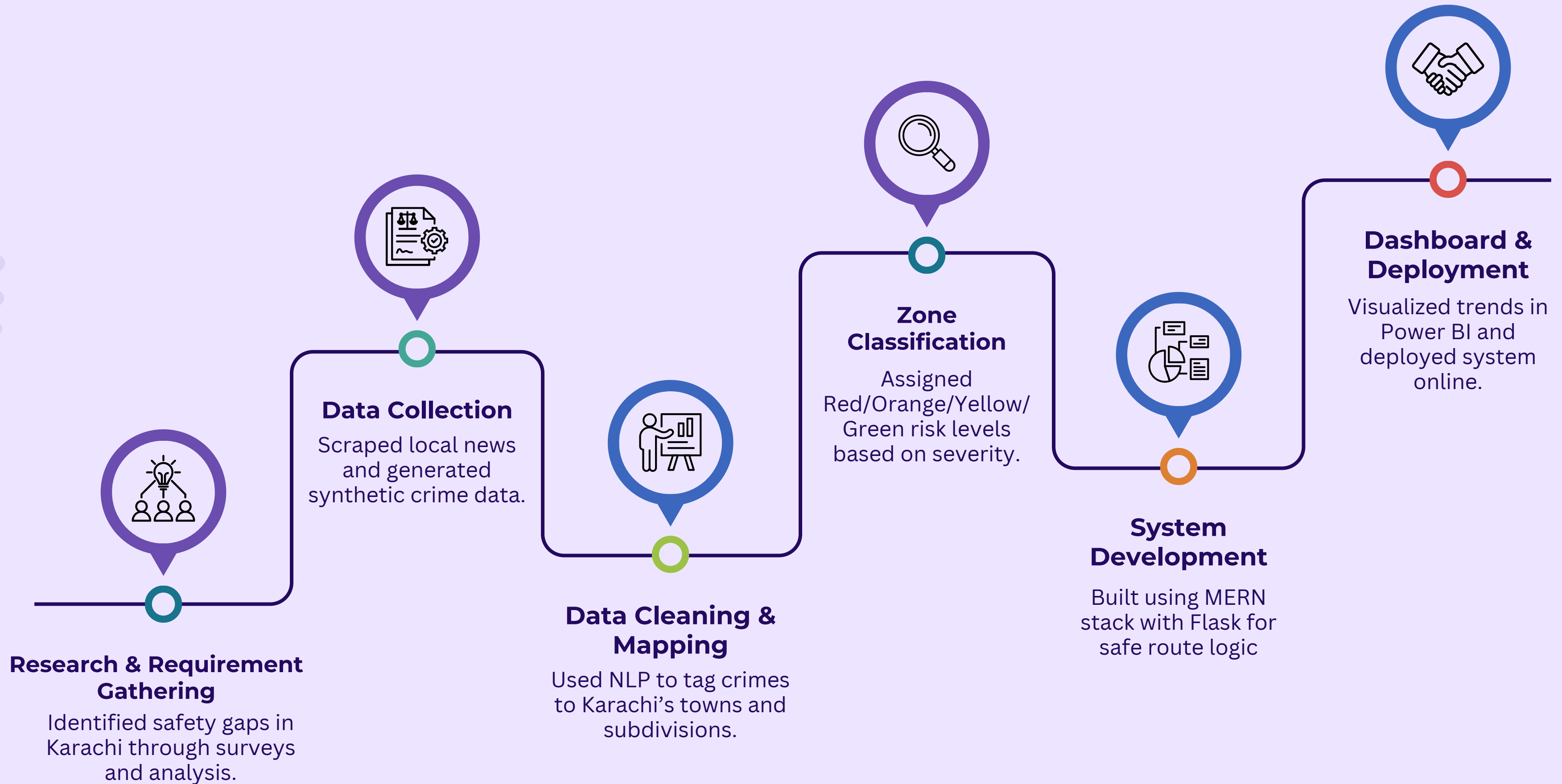
- Police data is delayed & unstructured
- Existing apps only offer SOS alerts
- No tool for:
- Live red zone mapping
- Risk-based safe route planning



Goals & Objectives

- **Red Zone Mapping** (area risk levels: Red, Orange, Yellow, Green)
- **Safe Route Suggestions** (risk-aware routing using Dijkstra algorithm)
- **Community Reporting** (users share local incidents)
- **Visual Dashboards** (crime trends via Power BI)
- **Safety Resources** – Police contacts, self-defense videos, legal support info

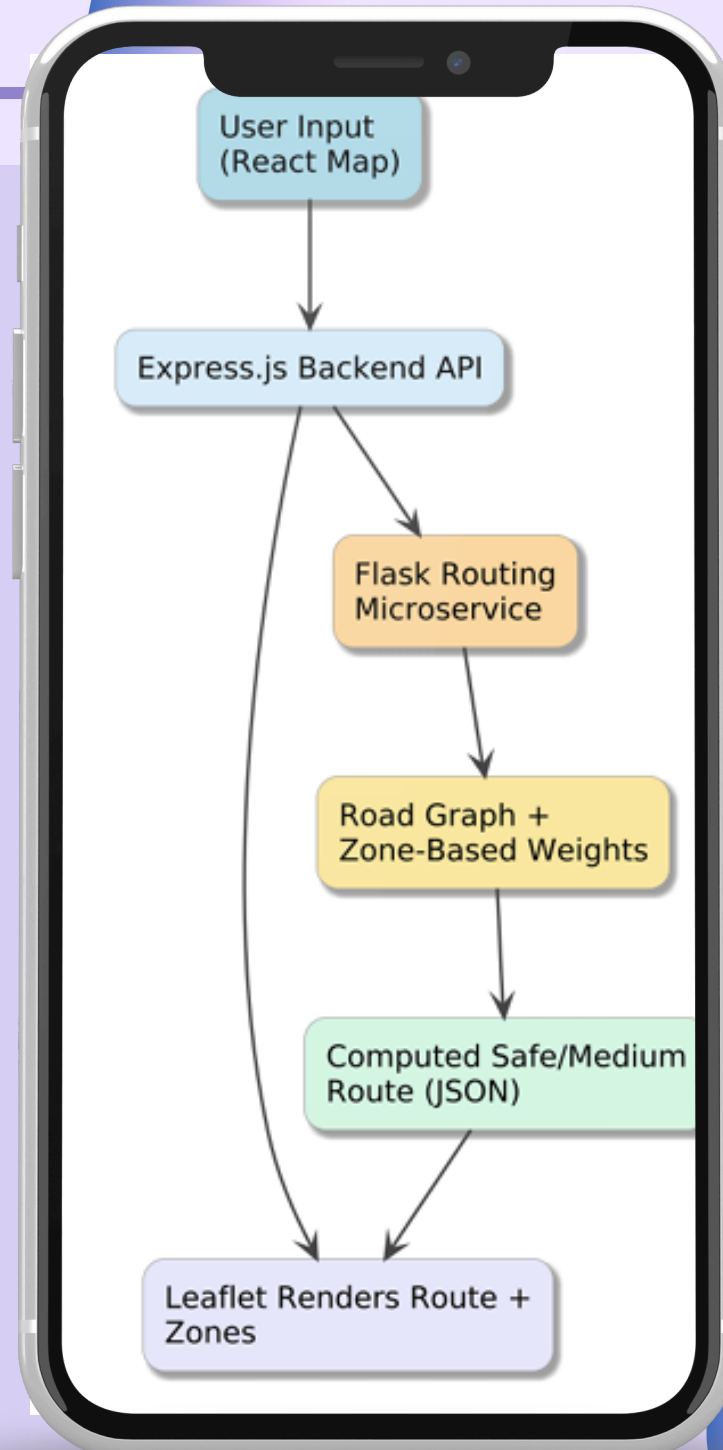




Methodology: Data Collection Process

● Primary Data Collection Attempts

- Approached 10+ NGOs & Police (Aurat Foundation, CPLC, WAR, etc.)
- No structured, shareable datasets were available



● Alternative Strategy

- Web scraping from news portals (Dawn, Geo, ARY)
- Used tools: BeautifulSoup, Selenium, Outscraper, Google CSE
- Extracted ~1800 usable incident reports

● Synthetic Dataset

- Created 100,000+ synthetic records using Faker.js
- Balanced real vs. underreported zones like Lyari, Korangi, etc.



G	H	I	J	K	L	M
SUBDIVISI	SEVERITY	SEVERITY	DATE	LATITUDE	LONGITUDE	CRIME
1	10	High	12/15/2024	24.8592	66.9998	Gang
1	10	High	12/3/2020	24.8592	66.9998	Murd
1	10	High	1/24/2025	24.8592	66.9998	Murd
1	10	High	2/18/2023	24.8592	66.9998	Gang
1	10	High	4/13/2025	24.8592	66.9998	Gang
1	10	High	4/14/2021	24.8592	66.9998	Gang
1	10	High	2/28/2021	24.8592	66.9998	Kidna
1	10	High	10/6/2020	24.8592	66.9998	Murd
1	10	High	12/3/2022	24.8592	66.9998	Murd
1	10	High	6/19/2022	24.8592	66.9998	Gang
1	10	High	8/17/2022	24.8592	66.9998	Shoot
1	10	High	12/12/2020	24.8592	66.9998	Murd
1	9	High	3/23/2023	24.8592	66.9998	Shoot
1	10	High	6/17/2020	24.8592	66.9998	Shoot
1	10	High	11/18/2024	24.8592	66.9998	Kidna
1	10	High	5/3/2024	24.8592	66.9998	Murd
1	10	High	8/22/2021	24.8592	66.9998	Gang
1	10	High	8/10/2020	24.8592	66.9998	Shoot
1	9	High	11/5/2022	24.8592	66.9998	Drug
1	9	High	12/2/2024	24.8592	66.9998	Sho



Data Processing & Risk Classification

- Location tagging using NLP (spaCy) + manual mapping
- Subdivision mapped to 18 Karachi towns
- Risk Classification: Red, Orange, Yellow, Green
- Output stored in karachi_crime_dataset_ranked.csv

Tech Stack & Architecture

- **Frontend:**

React + Vite (Leaflet for map)

- **Backend:**

Node.js + Express.js (APIs)

- **Python Microservices:**

Flask (Zone + Routing API)

- **Routing Engine:**

- Dijkstra algorithm
- Risk-weighted paths (Red = high weight)

- **Database:**

MongoDB Atlas

- **Security:**

- JWT auth
- Role-based access
- Rate limiting

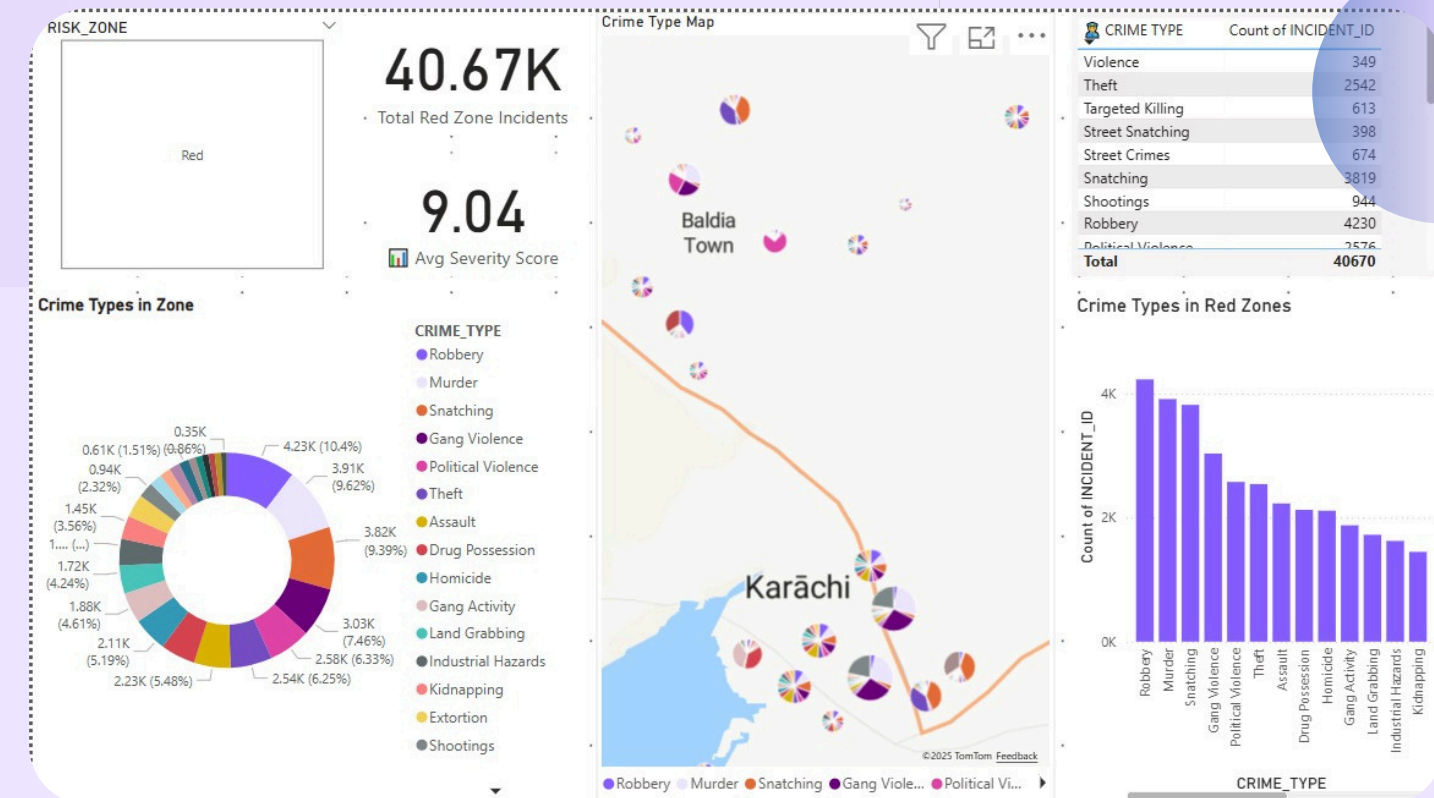
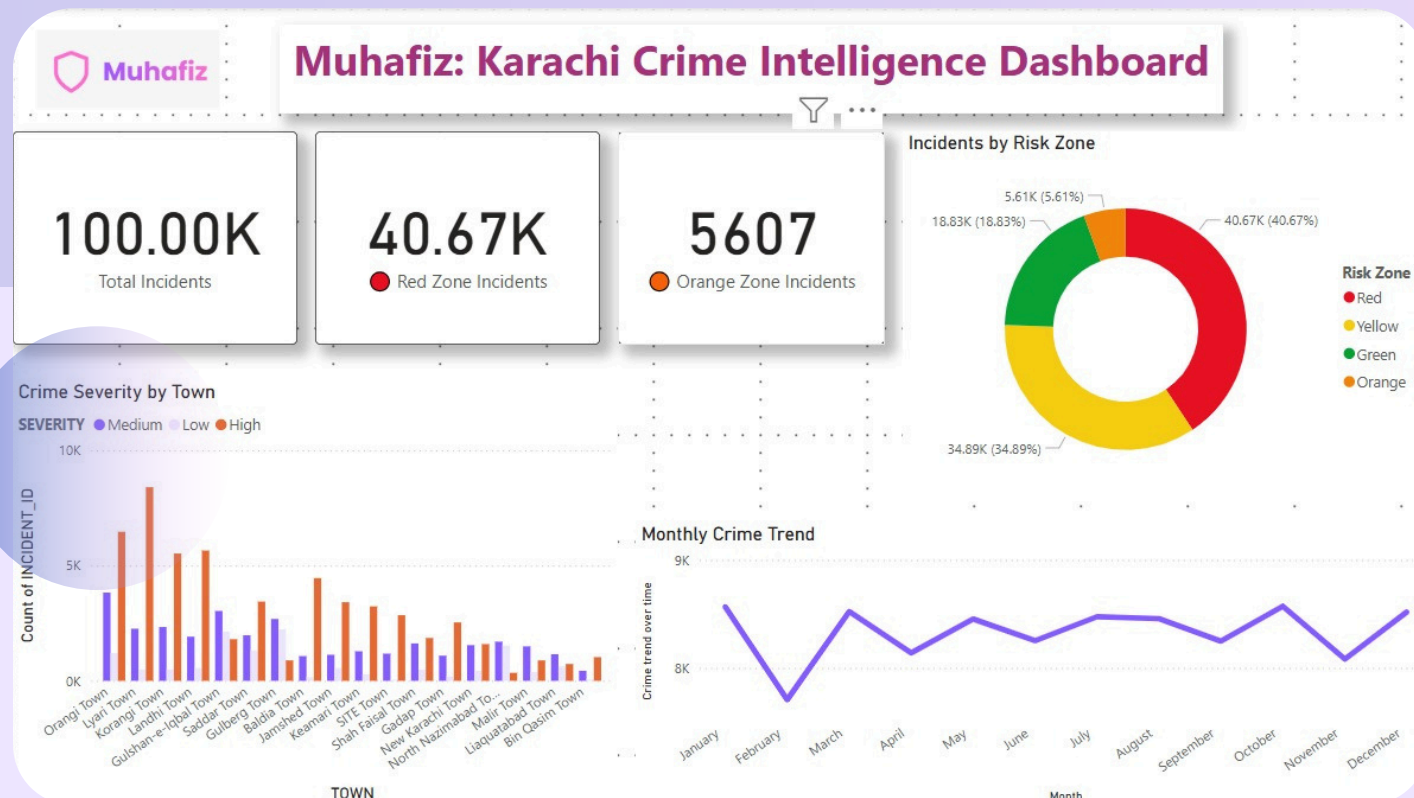
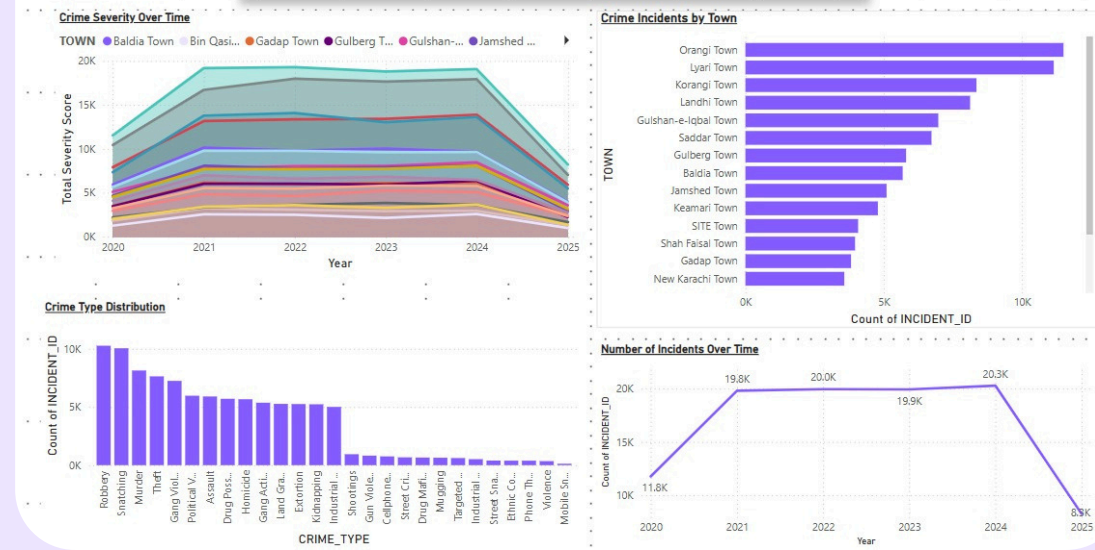
Power BI Dashboard

Purpose:

Visual insight for users and admins
Identifies hotspots & pattern changes

Tools: CSV integration with Power BI, conditional color coding for zones

Crime Trends Overview



Key Challenges & Solutions

Challenge	How We Solved It
Lack of public crime data	Created a large-scale synthetic dataset using local insights
Vague incident locations	Applied NLP + manual mapping for accurate geotagging
Police/NGO non-cooperation	Used web scraping + field validation
Privacy concerns	Applied JWT auth, role-based access
Tech stack integration	Linked Flask with MERN using secure REST APIs
User dependency	Added forums to drive engagement



Future Enhancements

Mobile App Launch

→ Native Android/iOS app for real-time mobile safety

Live Crime Prediction (ML)

→ Train ML models for real-time zone reclassification

Offline Functionality

→ Cached maps & alerts for users with low internet access

Trusted Authority Integration

→ Collaborate with police & NGOs for verified data input

Women & Children Mode

→ Add travel alerts + location-based family check-in

Voice-Based Emergency Reporting

→ Report incidents hands-free in emergencies



Thank You.



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